

TEST CASES OF HOLONOMIC CONSTRAINTS IN A FINITE ELEMENT ANALYSIS FRAMEWORK

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ABSTRACT. To come some day.

1. INTRODUCTION

1.1. Observations.

- (1) Yaw data is often suspect.
- (2) Differentiation often creates a noisy result, e.g., differentiate position to get velocity, in the computation of ejection speed, results in noisy “spray” either to the left or right (latitude) of the roll path (longitude).
- (3) Have experimented with shutting off (zero-out) yaw rates, but retain yaw angles, and has more smooth and (perhaps) more believable results.

1.2. **Questions.** Given a canonical rollover example, e.g., Exponent Expedition dolly rollover test:

- (1) What is the breakdown of ejection speed into components (directions: latitude and longitude) and into contribution source (e.g., $\omega \times r$, carrier frame speed, relative speed in rotating frame).
- (2) Is there a contribution source that is dominant for all time?
- (3) What is the *relative* contribution of each (e.g., normalize to 1.0, examine breakdown).
- (4) Is there a justification for shutting off yaw rate?
- (5) Is there a justification for modeling yaw rate as some smooth “canned” functional form, assumed *a priori*?

2. METHODS

2.1. **Reference Frames.** Fig. 1 illustrates the reference frames. Let the inertial reference frame $\mathcal{F}$ have unit vectors $[\hat{\mathbf{f}}_1, \hat{\mathbf{f}}_2, \hat{\mathbf{f}}_3]$ at origin point F . Unit vector $\hat{\mathbf{f}}_1$ points to the right of the page. Unit vector $\hat{\mathbf{f}}_2$ points the top of the page. Unit vector $\hat{\mathbf{f}}_3$ points out of the page, and is perpendicular to a level plane of vehicle travel. These unit vectors correspond to the easting, northing, and elevation vectors in a plan view survey coordinate system, respectively. All unit vectors used in this paper are dextral, orthonormal, ordered triads. We note that our coordinate system uses the “z” direction as up. This is different from the SAE convention, which uses the “z” direction as down.

Let the carrier frame $\mathcal{A}$ be attached to the vehicle’s center of mass at point V . The carrier frame has unit vectors $[\hat{\mathbf{a}}_1, \hat{\mathbf{a}}_2, \hat{\mathbf{a}}_3]$. Unit vector $\hat{\mathbf{a}}_1$ points toward the front of the vehicle. Unit vector $\hat{\mathbf{a}}_2$ points to the left of the vehicle, with the vehicle in a non-rolling configuration. Unit vector $\hat{\mathbf{a}}_3$ points to the roof of the vehicle, with the vehicle in a non-rolling configuration. The carrier frame translates with the vehicle frame, but is not attached to the vehicle. Therefore, the carrier frame does not rotate with roll θ of the vehicle with respect to the with the vehicle roll axis the $\hat{\mathbf{b}}_1$ (equivalently $\hat{\mathbf{a}}_1$). The carrier frame rotates with the yaw ψ of the vehicle with respect to the inertial frame along the vertical axis $\hat{\mathbf{f}}_3$ (equivalently $\hat{\mathbf{a}}_3$).

Let the vehicle frame $\mathcal{B}$ be fixed to the vehicle’s center of mass at point V . The vehicle frame has unit vectors $[\hat{\mathbf{b}}_1, \hat{\mathbf{b}}_2, \hat{\mathbf{b}}_3]$. Unit vector $\hat{\mathbf{b}}_1$ points to the front of the vehicle. Unit vector $\hat{\mathbf{b}}_2$ points to the left of the vehicle. Unit vector $\hat{\mathbf{b}}_3$ points to the roof of the vehicle.

2.2. **Occupant Description at Ejection.** In our previous work (Hovey 2008), we described the position vector of the occupant at ejection $\mathbf{r}^{FO} \in \mathbb{R}^3$

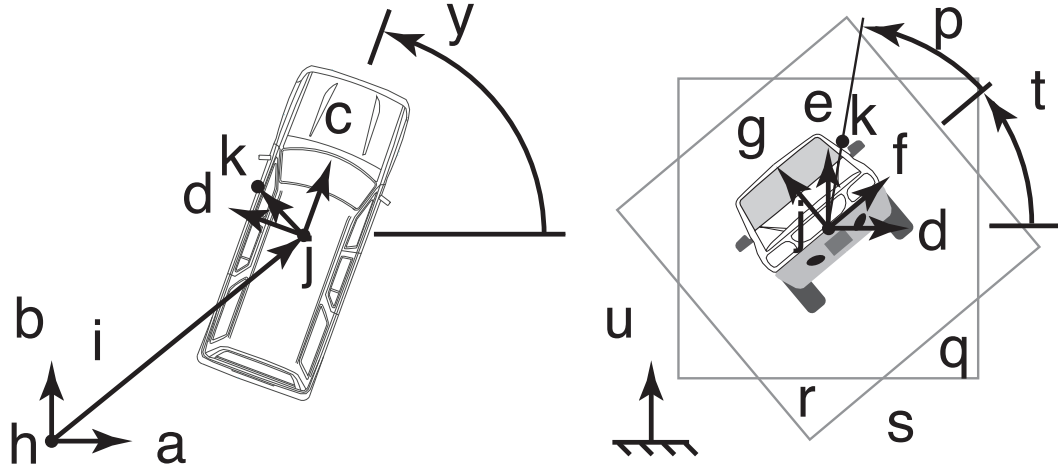


FIGURE 1. Illustration of inertial, carrier, and vehicle reference frames in a top view (left subfigure) and vehicle front view (right subfigure).

as

$$\mathbf{r}^{FO} = \mathbf{r}^{FV} + \mathbf{r}^{VO}, \quad (1)$$

$$\mathbf{r}^{FO} = x \hat{\mathbf{f}}_1 + y \hat{\mathbf{f}}_2 + z \hat{\mathbf{f}}_3 + \chi \hat{\mathbf{b}}_1 + \rho \cos \phi \hat{\mathbf{b}}_2 + \rho \sin \phi \hat{\mathbf{b}}_3. \quad (2)$$

We described the velocity of an occupant $\mathcal{F}_{\mathbf{v}^O} \in \mathbb{R}^3$ ejected from a rollover vehicle with non-trivial yaw effects as

$$\mathcal{F}_{\mathbf{v}^O} = \mathcal{F}_{\mathbf{v}^V} + \mathcal{B}_{\mathbf{v}^O} + \mathcal{F}_{\boldsymbol{\omega}^B} \times \mathbf{r}^{VO}, \quad (3)$$

where $\mathcal{F}_{\mathbf{v}^V}$ is the velocity of the vehicle relative to the inertial frame, $\mathcal{B}_{\mathbf{v}^O}$ is the velocity of the occupant relative to the vehicle frame, $\mathcal{F}_{\boldsymbol{\omega}^B}$ is the angular velocity of the vehicle in the inertial frame, and $\mathbf{r}^{VO}$ is the position vector from the vehicle at point V to the occupant at point O .

Writing this equation out explicitly in terms of velocity of the vehicle center of mass in the inertial reference frame $(\dot{x}, \dot{y}, \dot{z})$, the radial and longitudinal position (ρ, χ) and velocity $(\dot{\rho}, \dot{\chi})$ of the occupant in the vehicle reference frame, the yaw and roll position (ψ, θ) and rates $(\dot{\psi}, \dot{\theta})$, and using the shorthand notation of $\cos \alpha = c\alpha, \sin \alpha = s\alpha$ for all angle variables α gives

$$\mathcal{F}_{\mathbf{v}^O} = \begin{Bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \end{Bmatrix} + \begin{bmatrix} c\psi & -s\psi c\theta & s\psi s\theta \\ s\psi & c\psi c\theta & -c\psi s\theta \\ 0 & s\theta & c\theta \end{bmatrix} \begin{Bmatrix} \dot{\chi} \\ \dot{\rho} c\phi \\ \dot{\rho} s\phi \end{Bmatrix} +$$

$$\begin{bmatrix} 0 & -\dot{\psi} & s\psi \dot{\theta} \\ \dot{\psi} & 0 & -c\psi \dot{\theta} \\ -s\psi \dot{\theta} & c\psi \dot{\theta} & 0 \end{bmatrix} \begin{bmatrix} c\psi & -s\psi c\theta & s\psi s\theta \\ s\psi & c\psi c\theta & -c\psi s\theta \\ 0 & s\theta & c\theta \end{bmatrix} \begin{Bmatrix} \chi \\ \rho c\phi \\ \rho s\phi \end{Bmatrix} \quad (4)$$

which is equivalent to the result we recorded previously (Hovey 2008).

2.3. Specialization to Roll Dominant Condition.

Suppose now that the vehicle is predominantly in a roll, with trivial yaw. If we approximate this condition as the constant $\psi = -90^\circ + \epsilon$, where $\epsilon \approx 0$. This means $\dot{\psi} \neq 0$ for all time. We are then describing a vehicle that has yawed approximately one quarter of a revolution clockwise, when viewed from above, as it moves. Eq. (4) then simplifies to

$$\begin{aligned} \mathcal{F}_{\mathbf{v}^O}|_{\psi=-90^\circ} &= \begin{Bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \end{Bmatrix} + \begin{bmatrix} 0 & c\theta & -s\theta \\ -1 & 0 & 0 \\ 0 & s\theta & c\theta \end{bmatrix} \begin{Bmatrix} \dot{\chi} \\ \dot{\rho} c\phi \\ \dot{\rho} s\phi \end{Bmatrix} + \begin{Bmatrix} 0 \\ \dot{\psi} \\ -\dot{\theta} \end{Bmatrix} \\ &= \begin{Bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \end{Bmatrix} + \begin{Bmatrix} \dot{\rho} (c\theta c\phi - s\theta s\phi) \\ -\dot{\chi} \\ \dot{\rho} (s\theta c\phi + c\theta s\phi) \end{Bmatrix} + \begin{Bmatrix} 0 \\ \dot{\psi} \\ -\dot{\theta} \end{Bmatrix} \end{aligned}$$

The only contributions to the velocity in the $\mathcal{F}_{\mathbf{v}^O} \cdot \hat{\mathbf{f}}_2$ direction arise from vehicle speed $\dot{y}$, occupant longitudinal speed relative to the vehicle frame $\dot{\chi}$, and yaw rate $\dot{\psi}$. These contributions are linear. Perturbations in $\dot{y}$ alone cause one-to-one corresponding

perturbations in the occupant's ejection speed in the $\hat{\mathbf{f}}_2$ direction. Identical arguments can be made for $\dot{\chi}$ and $\dot{\psi}$. The significance of this development is to illustrate sensitivity of the occupant's $\hat{\mathbf{f}}_2$ ejection speed. If these values, $\dot{y}$, $\dot{\chi}$, and $\dot{\psi}$ are zero, the values for $\mathcal{F}_{\mathbf{v}^O}$ are forever confined to the plane spanned by $[\hat{\mathbf{f}}_1, \hat{\mathbf{f}}_3]$.

Next, we specialize the foregoing result for trailing and leading occupants. To facilitate insight without stretching the model too far from reality, we take the leading occupant ejection angle $\phi_{\text{lead}} = 0^\circ$ and the trailing occupant ejection angle $\phi_{\text{trail}} = 180^\circ$. Typically, when $\psi = -90^\circ$, we take $\phi_{\text{lead}} = 50^\circ$ and $\phi_{\text{trail}} = 145^\circ$. Also, for sunroof ejection portals, we typically start $\phi_{\text{roof}} = 90^\circ$. Thus, by taking a wide range $[0, 180]^\circ$, we are bounding the ejection parameter for ϕ below by 0° and above by 180° . Substitution of numeric values of ϕ produces the following two results:

$$\mathcal{F}_{\mathbf{v}^O} \Big|_{\phi=0^\circ}^{\psi=-90^\circ} = \begin{Bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \end{Bmatrix} + \begin{Bmatrix} \dot{\rho} c\theta \\ -\dot{\chi} \\ \dot{\rho} s\theta \end{Bmatrix} + \begin{Bmatrix} \rho s\theta \dot{\theta} + \chi \dot{\psi} \\ \rho c\theta \dot{\psi} \\ \rho c\theta \dot{\theta} \end{Bmatrix} \quad (7)$$

$$\mathcal{F}_{\mathbf{v}^O} \Big|_{\phi=180^\circ}^{\psi=-90^\circ} = \begin{Bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \end{Bmatrix} + \begin{Bmatrix} -\dot{\rho} c\theta \\ -\dot{\chi} \\ -\dot{\rho} s\theta \end{Bmatrix} + \begin{Bmatrix} -\rho s\theta \dot{\theta} + \chi \dot{\psi} \\ -\rho c\theta \dot{\psi} \\ -\rho c\theta \dot{\theta} \end{Bmatrix} \quad (8)$$

3. RESULTS

3.1. Special Case: 2D Ejection Solutions with Bounded Ejection Angles.

3.2. Special Case: 2D Ejection Solutions with Tuned Ejection Angles.

3.3. 3D Ejection Solutions with Naive Yaw Data.

3.4. 3D Ejection Solutions with Damped Yaw Data.

3.5. 3D Ejection Solutions with Smoothed Yaw Data.

4. DISCUSSION

5. CONCLUSIONS

6. REFERENCES

APPENDIX A. NUMERICAL DETAILS

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